23-XX-9105
1ST STAGE SUCTION/DISCHARGE COOLER SKID

23-HA-9114/9115
1st STAGE DISCHARGE COOLERS
DUTY: 15330 kW
TYPE: SHELL AND TUBE
DESIGN SHELL: 15.8 BARG/FV @ 150 °C
OPERATING SHELL: 5.0 BARG @ 20.12°C
DESIGN TUBE: 30 BARG/FULL VACUUM AT 75°C (-28°C MIN)
OPERATING TUBE: 12.0 BARG AT 9.7°C
MATERIAL: 22 Cr DUPLEX

HP FLARE HEADER
PH-ME-P-0268-002
LP FLARE HEADER
PH-ME-P-0270-001
FROM 1st STAGE COMPRESSOR
PH-ME-P-0153-001
2400-PV-23L91377-BD20-7C
2400-PV-23L91381-BD20-7C
0400-VF-31L92648-AS23-0
0400-VF-31L92481-AS23-0
0300-VF-31L92647-AD20-0
0400-WC-45L94556-AD23-0
0300-WC-45L94433-AD20-0
0300-WC-45L94424-AD20-0
0600-WC-45L94451-AD20-1C
HOT COOLING MEDIUM RETURN
PH-ME-P-0265-001
0200-WC-45L94453-AD20-1C
0600-WC-45L94450-AD20-1C
23-HA-9115
TUBE: 0200-PV-23L91447-BD20-3
SHELL: 0200-WC-45L94487-AD20-0
23-HA-9114
TUBE: 0200-PV-23L91448-BD20-3
0600-WC-45L94447-AS20-0
3000-PV-23L91386-AD20-7C
SURGE CONTROL AND RECYCLE TO 1ST STAGE SUCTION SCRUBBER
PH-ME-P-0152-001
0300-VF-31L92646-AD20-0
1800-PV-23L91384-BD20-0
0200-PV-23L91385-BD20-0
PDT MEASUREMENT FROM 23-V-90367
PH-ME-P-0161-001
TO 2nd STAGE SUCTION SCRUBBER
PH-ME-P-0161-001
2400-PV-23L91383-BD20-7C
23-XX-9105
SKID LIMITS
COOLING MEDIUM SUPPLY
PH-ME-P-0265-001
0600-WC-45L94390-AS20-0
0200-DO-81L96504-AD20-0
0200-PV-23L91423-BD20-0
0200-DO-81L96664-AD20-0
HAZARDOUS OPEN DRAIN
PH-ME-P-0280-002
0200-DO-81L96644-AD20-0
0200-DC-81L96505-BD20-0
CLOSED DRAIN
PH-ME-P-0278-001
CONT. ON DWG. PH-ME-P-0152-001
CONT. ON DWG. PH-ME-P-0161-001

NOTES:
1. KEY INTERLOCKED VALVE.
2. REMOVABLE MAINTANCE SPOOL.
3. CONDITIONING PLATE. PROVIDE STRAIGHT RUN OF 2D UPSTREAM AND 2D DOWNSTREAM.
4. KEY LOCKED VALVE.
5. MINIMUM 1 METER SEPARATION.
6. DO NOT POCKET.
7. LOCATE ANTI-SURGE VALVE AS CLOSE AS PRACTICAL TO TEE.
8. 2" EQUALIZATION VALVE PARALLEL TO SHUTDOWN VALVE.
9. DISSIMILAR CHECK VALVES.
10. TRANSMITTER TO BE MOUNTED OFF SKID ON WEATHER DECK DUE TO THE ELEVATION OF THE PDT TAPS ON SKID.
11. SEE DETAIL F101 ON P&ID PH-ME-P-0102-001.
12. PERMISSIVE TO OPEN LARGE ESDV VALVE ON LOW DIFFERENTIAL.
13. SIGNAL DIRECT TO PSD, REPEATED TO PCS SYSTEM.
14. INSTRUMENT LINES ARE HEAT TRACED.

REV	DATE	BY	DESCRIPTION	CHK	ENGR	APPR	CLIENT
Z3	07.04.16	[redacted]	AS BUILT, MW-33788				
Z2	28.01.15		AS BUILT, AFA V-4426				
Z1	26.06.14		AS BUILT, MW-19805				
Z	24.06.13		AS BUILT				

Wood Group Engineering (North Sea) Ltd.
MUSTANG INTERNATIONAL, LP
HOUSTON, TEXAS
PROJECT NO. 10910

bp

VALHALL
RE-DEVELOPMENT PROJECT
PIPING & INSTRUMENT DIAGRAM
1st STAGE DISCHARGE COOLER

REGISTRATION CODES			DRAWING NO.					SCALE
AREA	SYSTEM	TYPE	FACILITY CODE	ORIG.	DISP.	SEQ. No.	SHEET	NONE
T33-02	23	XB	PH	ME	P	0160	001	REV. Z3

Field for file specification:
Field for date and time :